HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

F79SP

STOCHASTIC PROCESSES

Semester 1— 2022/23

Duration: Two Hours

There are 5 questions in this paper, 80 marks in total.
AMS formula sheet is provided at the end of this paper.

1. (a) Let $S_n = a + \sum_{i=1}^{n} \xi_i$, $n = 0, 1, 2, \ldots$ be a simple random walk with increments having distribution $\mathbf{P}(\xi_i = 1) = p$ and $\mathbf{P}(\xi_i = -1) = q = 1 - p$ where $0 < p < 1$.

i) Show that the random walk forms a time-homogeneous Markov chain on the integers that is also space-homogeneous. [3]

ii) Is this Markov chain periodic or not? Justify your answer. [2]

iii) Using Stirling's formula, show that, as $n \to \infty$,

$$p_{00}^{(2n)} = \mathbf{P}(S_{2n} = 0 \mid S_0 = 0) \sim \frac{(4pq)^n}{\sqrt{\pi n}}.$$

Using this result or any other method, show that the Markov chain is recurrent if and only if $p = 1/2$. [4]

(b) Let $S_n = a + \sum_{i=1}^{n} \xi_i$, $n = 0, 1, 2, \ldots$ be a simple random walk with increments having distribution $\mathbf{P}(\xi_i = 1) = p > 0$, $\mathbf{P}(\xi_i = 0) = r \geq 0$ and $\mathbf{P}(\xi_i = -1) = q > 0$ where $p + r + q = 1$.

i) Find probabilities $\mathbf{P}$(there exists n such that $S_n \leq -10 \mid S_0 = 0)$ and $\mathbf{P}$(there exists n such that $S_n \geq 10 \mid S_0 = 0)$ and justify that at least one of these probabilities equals 1. Further, when are both these probabilities equal to 1? [4]

ii) Assume now that $p = 2/3$, $r = 0$ and $q = 1/3$. Let $T_1 = \min\{n \geq 1 : \ S_n \leq -10 \mid S_0 = 0)$ and $T_2 = \min\{n \geq 1 : \ S_n \geq 10 \mid S_0 = 0)$. Find the values of $\mathbf{E}T_1$ and $\mathbf{E}T_2$ in this case. [3]

[Total 16 Marks]

2. A Markov chain $\{X_n\}_{n\geq 0}$ takes values in the set $\{1,2,3,4,5,6,7\}$ and has transition matrix

$$\begin{pmatrix} 1/4 & 0 & 0 & 1/4 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 & 0 & 0 & 0 & 0 \\ 1/4 & 0 & 0 & 1/4 & 1/2 & 0 & 0 \\ 1/2 & 0 & 0 & 1/4 & 1/4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1/4 & 0 & 1/4 & 0 & 1/4 & 1/4 \end{pmatrix}$$

(a) Identify the open and closed classes, and justify your answers. [2]

(b) Determine the limiting distribution if the Markov chain starts from the initial state $X_0 = 1$. [4]

(c) Find all stationary distributions of the Markov chain. [3]

(d) Determine the limiting distribution if the Markov chain starts from the initial state $X_0 = 3$. [5]

(e) What is the probability that the Markov chain ever visits state 7 given that it starts from the initial state $X_0 = 3$? [3]

(f) What is the mean number of jumps for the Markov chain to get to any closed class, given that it starts from state $X_0 = 2$? [4]

(g) Assume that the Markov chain starts from the initial distribution $\mathbf{P}(X_0 = i) = 1/3$, for $i = 2,3,4$. Find the conditional probability

$$\mathbf{P}(X_0 = 2 \mid X_2 = 4).$$

[3]

[Total 24 Marks]

3. Consider a discrete-time Galton-Watson branching process $\{X_n, n \geq 0\}$. Let p_i denote the probability that an individual has i offspring, $i = 0, 1, 2, \ldots$. We assume that

$$p_0 = p_1 = p_2 = p_3 = 1/4 \quad \text{and} \quad p_i = 0 \text{ for } i \geq 4.$$

Let $\tau_k(S)$ be the time to hit the set $S = \{0, 4, 5, 6, 7, 8, 9, 10\}$ for the process with initial state $X_0 = k$, i.e.

$$\tau_k(S) = \min\{n \geq 1 : X_n \in S \mid X_0 = k\}.$$

(a) Find the transition probabilities p_{ij} of the Markov process $\{X_n\}$ for $i, j = 1, 2, 3$. [4]

(b) Find the probability $\mathbf{P}(\tau_1(S) \geq 3)$. [2]

(c) Derive the system of linear equations for $\mathbf{E}\tau_1(S)$, $\mathbf{E}\tau_2(S)$ and $\mathbf{E}\tau_3(S)$. (You are not supposed to solve it). [2]

[Total 8 Marks]

4. **(a)** Let ψ_1, ψ_2, ψ_3 be three mutually independent random variables and assume that, for $i = 1, 2, 3$, random variable ψ_i has Exponential distribution with parameter $\alpha_i > 0$.

i) Prove that random variable ψ_1 has the memoryless property: for any $t, u > 0$,

$$\mathbf{P}(\psi_1 > t + u \mid \psi_1 > t) = \mathbf{P}(\psi_1 > u).$$

[1]

ii) Find the distribution of the random variable $\min(\psi_1, \psi_2, \psi_3)$ and justify your answer. [2]

(b) For $t \geq 0$, let $N_1(t), N_2(t), N_3(t)$ be three mutually independent time-homogeneous Poisson processes, with corresponding intensities λ_1, λ_2 and λ_3. Let $N(t)$ be the superposition of these 3 processes.

i) Prove that $N(t)$ is also a time-homogeneous Poisson process and find its intensity.

[4]

ii) Find the distribution of the first point of the process $N(t)$. [2]

iii) Find the distribution of the second point of the process $N(t)$. [2]

(c) For a certain insurance company, we model the process of claim arrivals $N(t)$ as a *time-inhomogeneous* Poisson process with intensity function $\lambda(t)$. We assume that time is measured in hours and that, within every 24 hours (from midnight to the next midnight), $\lambda(t) = 2$ between midnight and 6am and $\lambda(t) = 8$, otherwise.

i) For a given day, what is the conditional probability that the number of claims between 6am and 12pm is 4, given that the number of claims between 4am and 10am is 2? [5]

ii) Assume that the claim sizes are random, mutually independent and do not depend on the process $N(t)$, and that each of them takes large value $\pounds 6$, with probability 1/3 or small value $\pounds 3$, with probability 2/3.
What is the expected value of the total claim amount within a day? [2]
What is the probability that, within a day, there are at most 2 large claims, given that the total number of claims is 5? [3]

[Total 21 Marks]

5. A continuous-time homogeneous Markov process $\{X(t)\}_{n\geq 0}$ takes values in the set $\{1,2,3,4,5\}$ and has a transition rate matrix

$$Q = \begin{pmatrix} -3 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 \\ 4 & 0 & 0 & -4 & 0 \\ 1 & 1 & 0 & 1 & -3 \end{pmatrix}.$$

(a) Find the transition matrix of the embedded Markov chain. [2]

(b) Classify the states of the Markov process and justify the classification. [2]

(c) Do all stationary distributions of the Markov process and of its embedded Markov chain coincide or not? [2]

(d) Assume that the initial value of the Markov process is non-random, $X(0) = i \in \{1,2,3,4,5\}$. For which values of i, do there exist both the limiting distribution of the Markov process and the limiting distribution of the embedded Markov chain? [5]

[Total 11 Marks]

END OF PAPER

Some Absolutely Continuous Distributions

Name	Notation	p.d.f.	range	m.g.f.	mean	variance
Uniform	$U(a,b)$	$\frac{1}{b-a}$	$a \le x \le b$	$\frac{e^{b\theta}-e^{a\theta}}{(b-a)\theta}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	$N(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$	$-\infty < x < \infty$	$e^{\mu\theta+\frac{1}{2}\sigma^2\theta^2}$	μ	σ^2
Gamma	$\Gamma(t,\lambda)$	$\frac{\lambda^t}{\Gamma(t)}x^{t-1}e^{-\lambda x}$	$0<x<\infty$, $t,\lambda>0$	$\left(\frac{\lambda}{\lambda-\theta}\right)^t$	$\frac{t}{\lambda}$	$\frac{t}{\lambda^2}$
Exponential	$\mathrm{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$0<x<\infty$, $\lambda>0$	$\frac{\lambda}{\lambda-\theta}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Chi-square	χ^2_n	$\frac{1}{2^{n/2}\Gamma(n/2)}x^{n/2-1}e^{-x/2}$	$0<x<\infty$, $n>0$ integer	$(1-2\theta)^{-n/2}$	n	$2n$
Beta	$\beta(\alpha_1,\alpha_2)$	$\frac{\Gamma(\alpha_1+\alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)}x^{\alpha_1-1}(1-x)^{\alpha_2-1}$	$0<x<1$, $\alpha_1,\alpha_2>0$	not useful	$\frac{\alpha_1}{\alpha_1+\alpha_2}$	$\frac{\alpha_1\alpha_2}{(\alpha_1+\alpha_2)^2(\alpha_1+\alpha_2+1)}$
Cauchy	–	$\frac{\lambda}{\pi(\lambda^2+(x-\mu)^2)}$	$-\infty<x<\infty$, $\lambda>0$	doesn't exist but char. fn. is $e^{i\mu\theta-\lambda\lvert\theta\rvert}$	no moments exist	
Multivariate Normal	$N(\mu,\Sigma)$	$\frac{\exp\left\{-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right\}}{(2\pi)^{d/2}\det(\Sigma)^{1/2}}$	$x\in\mathbb{R}^d$	$e^{\mu^T\theta+\frac{1}{2}\theta^T\Sigma\theta}$	μ	covariance matrix Σ

Some Discrete Distributions

(All distributions below have integer-valued ranges)

Name	Notation	prob. fn.	range	m.g.f.	mean	variance
Binomial	$\mathrm{Bin}(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	$0\le x\le n$ $0<p<1$	$[(1-p)+pe^\theta]^n$	np	$np(1-p)$
Poisson	$\mathrm{Pois}(\lambda)$	$\frac{e^{-\lambda}\lambda^x}{x!}$	$0\le x<\infty$ $\lambda>0$ real	$e^{\lambda(e^\theta-1)}$	λ	λ
Hyper-geometric	$\mathrm{Hgeo}(n,a,b)$	$\frac{\binom{a}{x}\binom{b}{n-x}}{\binom{a+b}{n}}$	$0\le x\le n$ $a,b>0$ integers	not useful	$\frac{na}{a+b}$	$\frac{nab(a+b-n)}{(a+b)^2(a+b-1)}$
Negative binomial	$\mathrm{Nbin}(k,p)$	$\binom{x-1}{k-1}p^k(1-p)^{x-k}$	$k\le x<\infty$, $0<p<1$	$p^k[e^{-\theta}-(1-p)]^{-k}$	k/p	$k(1-p)/p^2$
Geometric	$\mathrm{Geo}(p)$	$p(1-p)^{x-1}$	$1\le x<\infty$ $0<p<1$	$p[e^{-\theta}-(1-p)]^{-1}$	$1/p$	$(1-p)/p^2$

Multinomial: multivariate extension of binomial

Prob. fn. is given by

$$f(x_1,x_2,\ldots,x_k)=\frac{n!}{x_!x_2!\ldots x_k!x_{k+1}!}p_1^{x_1}p_2^{x_2}\ldots p_k^{x_k}p_{k+1}^{x_{k+1}},\quad 0\le x_i\le n,\ \ 0<p_i<1\ \ \forall i=1,2,\ldots,n+1$$

where $p_1+p_2+\cdots+p_{k+1}=1$ and $x_1+x_2+\cdots+x_{k+1}=n$.

The m.g.f. is given by

$$[p_1e^{\theta_1}+p_2e^{\theta_2}+\cdots+p_ke^{\theta_k}+p_{k+1}]^n.$$